RGPV B.Tech 1st Year (AI/ML) — 2026

MOST-PROBABLE QUESTIONS (Unit-wise probability analysis)

Based on actual papers Nov 2022 → Jun 2026 · Generated 12 Oct 2026 · unofficial study material

Legend: 🔥 SURE-SHOT 82%+ · ⭐ HIGH 68-81% · 📘 MEDIUM 55-69% · ○ SLEEPER (rotation). Probability = frequency + recency + streak weight (2022-2026 papers se).

BT-201 — Engineering Physics

Unit	Question / Topic	Sessions	Prob.	Level
U1	Time-independent Schrödinger equation derivation	Jun2020, Jun2022, Dec2022, Dec2025, Jun2026	**93%**	🔥 SURE-SHOT
U1	Heisenberg uncertainty principle (+ gamma-ray microscope derivation)	Nov2018, May2019, Jun2022, Dec2022, Dec2025	**87%**	🔥 SURE-SHOT
U2	Plane transmission grating — theory, resolving power, dispersive power	2015, Jun2017, Nov2018, Dec2022, Dec2025	**85%**	🔥 SURE-SHOT
U2	Newton's rings — formation, fringe diameters, wavelength/RI numerical	Dec2020, May2018, Jun2022, Dec2022, Dec2025	**84%**	🔥 SURE-SHOT
U1	Particle in one-dimensional box — energy eigenvalues + wavefunction + numerical	Jun2020, Jun2022, Dec2025, Jun2026	**81%**	⭐ HIGH
U4 (Lasers & Fibre Optics)	Optical fibre — acceptance angle, acceptance cone, NA (+ numerical)	Jun2013, Nov2018, Dec2025, Jun2026	**78%**	⭐ HIGH
U2	Michelson interferometer — construction, working, fringe shift	May2019, Dec2020, Jun2022, Dec2025	**71%**	⭐ HIGH
U1	Relation between phase velocity and group velocity	Jun2020, Dec2022, Dec2025	**69%**	⭐ HIGH
U1	de Broglie hypothesis of matter waves / wave nature of particles	Dec2010, Jun2015, Dec2025	**65%**	📘 MEDIUM
U2	Rayleigh criterion of resolution / resolving power	2007, Nov2018, Dec2025	**65%**	📘 MEDIUM
U2	Fraunhofer diffraction at single slit (+ 4.5% subsidiary maximum)	Jun2020, Dec2025	**60%**	📘 MEDIUM
U1	Physical significance of wave function / Born interpretation	Jun2022, Dec2022	**32%**	○ SLEEPER (rotation)
U2	Mach–Zehnder interferometer	Dec2022	**32%**	○ SLEEPER (rotation)
U1	Compton scattering — Compton shift derivation	Dec2016, 2017, May2018, Nov2018	**29%**	○ SLEEPER (rotation)
U2	Fresnel's biprism / division of amplitude vs wavefront	Jun2010, Dec2010, Jun2017, May2018	**29%**	○ SLEEPER (rotation)

BT-202 — Mathematics-II

Unit	Question / Topic	Sessions	Prob.	Level
U2	Method of variation of parameters	Jun2025, Dec2025, Jun2026	**86%**	SURE-SHOT
U2	Cauchy–Euler / reducible to homogeneous form	Jun2025, Dec2025, Jun2026	**86%**	SURE-SHOT
U1	Linear differential equation of first order (integrating factor)	Jun2025, Dec2025, Jun2026	**82%**	SURE-SHOT
U1	Homogeneous / variable separable first-order equation	Jun2025, Jun2026	**69%**	HIGH
U2	Complementary function + Particular integral (operator method)	Jun2025, Jun2026	**69%**	HIGH
U2	Series solution — Frobenius method	Jun2025, Dec2025	**67%**	MEDIUM
U1	Bernoulli's differential equation	Jun2026	**60%**	MEDIUM
U2	Bessel's function identity (J½(x) etc.)	Jun2026	**54%**	∘ SLEEPER (rotation)
U2	Simultaneous differential equations (dx/dt, dy/dt)	Dec2025	**49%**	∘ SLEEPER (rotation)
U2	Legendre's equation solved in series	Jun2025	**46%**	∘ SLEEPER (rotation)
U1	Exact differential equations / integrating factors; first order but not first degree	older	**28%**	∘ SLEEPER (rotation)
U2	Total differential equations	older	**28%**	∘ SLEEPER (rotation)

BT-203 — Basic Mechanical Engineering

Unit	Question / Topic	Sessions	Prob.	Level
U1	Iron–Carbon / Fe–Fe_3C equilibrium diagram — phases, invariant reactions, critical points	Nov2022, Jun2024, Dec2025, Jun2026	**92%**	SURE-SHOT
U2	Lathe machine — main parts + operations (with neat diagram)	Nov2022, Jun2024, Dec2025, Jun2026	**92%**	SURE-SHOT
U2	Temperature measurement instruments + optical pyrometer working	Nov2022, Jun2024, Dec2025, Jun2026	**92%**	SURE-SHOT
U1	Alloy steel / alloying elements ka effect; cast iron composition & classification	Nov2022, Dec2025, Jun2026	**82%**	SURE-SHOT
U1	Stress–strain curve of ductile material — yield point, UTS (+ numerical)	Nov2022, Jun2024, Dec2025	**75%**	HIGH
U1	Mechanical properties — strength, hardness, toughness, ductility, brittleness, malleability	Nov2022, Jun2024, Dec2025	**74%**	HIGH
U2	Vernier calliper — construction, least count, applications	Nov2022, Jun2026	**69%**	HIGH

Unit	Question / Topic	Sessions	Prob.	Level
U1	Hardness measuring methods; classification of engineering materials; alloys	Nov2022, Jun2026	**64%**	MEDIUM
U1	Viscosity — definition, importance + shear force/power numerical	Nov2022, Dec2025	**64%**	MEDIUM
U1	Hooke's law + modulus of elasticity derivation	Jun2026	**59%**	MEDIUM
U2	Slip gauge, sine-bar, dial gauge, combination set — short notes	Nov2022, Jun2024	**51%**	∘ SLEEPER (rotation)
U2	Errors in measurement (types) / measurement process	Jun2024	**41%**	∘ SLEEPER (rotation)
U2	Manometer / pressure measurement; absolute vs gauge pressure	Nov2022	**32%**	∘ SLEEPER (rotation)
U2	Welding defects	Nov2022	**32%**	∘ SLEEPER (rotation)

BT-204 — Basic Civil Engg. & Mechanics

Unit	Question / Topic	Sessions	Prob.	Level
U1	Bricks & stones — types, properties, tests, engineering uses	Nov2022, Jun2025, Jun2026	**82%**	SURE-SHOT
U2	Remote sensing — principle & civil engineering applications	Nov2022, Jun2025, Jun2026	**82%**	SURE-SHOT
U1	Cement — properties, laboratory tests, applications, concrete quality	Nov2022, Jun2026	**80%**	HIGH
U2	Levelling numerical — Height of Instrument / Rise & Fall method + arithmetic check	Nov2022, Jun2026	**80%**	HIGH
U2	Theodolite — components, principle, applications (+ electronic instruments)	Nov2022, Jun2026	**78%**	HIGH
U2	Contouring — principle, contour interval, uses; area & volume computation	Nov2022, Jun2026	**74%**	HIGH
U1	Stone masonry — types with sketches	Nov2022, Jun2025	**64%**	MEDIUM
U2	Plane table / survey stations / reciprocal levelling	Nov2022, Jun2026	**64%**	MEDIUM
U1	Foundations — spread footing & RCC footing (construction + suitability)	Jun2026	**62%**	MEDIUM
U1	Staircases / doors / windows — types & suitability	Nov2022, Jun2025	**62%**	MEDIUM
U1	Mortar — types + requirement estimation; concrete ke properties	Nov2022, Jun2025	**62%**	MEDIUM
U1	Plastering and pointing	Jun2026	**59%**	MEDIUM

Unit	Question / Topic	Sessions	Prob.	Level
U2	Measurement of distances & elevations; conventional vs EDM	Jun2026	**54%**	∘ SLEEPER (rotation)
U1	Brick masonry construction ke general principles	Jun2025	**46%**	∘ SLEEPER (rotation)
U2	GPS positioning principle	Jun2025	**46%**	∘ SLEEPER (rotation)

BT-205 — Basic Computer Engineering

Unit	Question / Topic	Sessions	Prob.	Level
U2	OOP characteristics / POP vs OOP comparison	Dec2022, Jun2024, Dec2025, Jun2026	**94%**	SURE-SHOT
U1	Operating system — definition, functions, file/process/memory management	Jun2020, Dec2022, Dec2025, Jun2026	**89%**	SURE-SHOT
U2	Algorithm — characteristics + flowchart (largest of three numbers) + program design	Jun2020, Jun2022, Jun2024, Dec2025, Jun2026	**88%**	SURE-SHOT
U1	Classification of computers + generations	Jun2011, Dec2012, Dec2022, Jun2024, Dec2025	**83%**	SURE-SHOT
U2	C++ program structure, tokens, arrays & functions	Jun2020, Jun2024, Dec2025, Jun2026	**83%**	SURE-SHOT
U2	Polymorphism — function/operator overloading, virtual functions	Dec2022, Jun2024, Jun2026	**82%**	SURE-SHOT
U1	CPU organization / block diagram of computer (registers, bus, instruction set)	Nov2018, Jun2020, Jun2024, Dec2025	**79%**	HIGH
U1	Memory & storage systems + I/O devices / memory hierarchy	Nov2018, Dec2025, Jun2026	**79%**	HIGH
U2	Constructors & destructors; friend function	Dec2017, Jun2022, Dec2025, Jun2026	**78%**	HIGH
U2	Inheritance — types with diagrams/examples; single vs multiple	Dec2017, Jun2022, Jun2024, Jun2026	**75%**	HIGH
U2	Scope resolution operator + local vs global variables	Dec2025, Jun2026	**73%**	HIGH
U1	MS-Word / MS-Excel / MS-PowerPoint features + office automation	Jun2024, Jun2026	**72%**	HIGH
U1	System software vs application software	Jun2011, Jun2020, Dec2025	**67%**	MEDIUM
U1	Computer applications — e-business, bioinformatics, healthcare, RS & GIS, multimedia	Dec2020, Jun2026	**63%**	MEDIUM
U1	Instruction set + steps CPU follows to execute an instruction	Dec2015, May2018, Jun2020	**35%**	∘ SLEEPER (rotation)

Unit	Question / Topic	Sessions	Prob.	Level
U2	Looping statements + programs (binary→decimal etc.)	Jun2020, Jun2022	**31%**	∘ SLEEPER (rotation)

Predicted Paper — BT-201 Engineering Physics (Mid-Sem / MSE-1)

Nov2022→Jun2026 ke 8 papers · attempt any five · 14 marks each

1. □ (a) State Heisenberg's uncertainty principle and derive it for the hypothetical gamma-ray microscope.

□ (b) Deduce the relation between phase velocity and group velocity for a free particle.

2. □ (a) Derive the time-independent Schrödinger wave equation and obtain the wave function for a particle in a one-dimensional potential box.

□ (b) Numerical: mass 9.11×10^{-31} kg, box width 10 nm — first three excited states ki energy nikaalo.

3. □ (a) Newton's rings — formation explain karo aur dark/bright fringe diameters ke expressions derive karo.

□ (b) Numerical: 4th & 12th dark ring ke diameters 0.400 cm & 0.700 cm → 20th dark ring ka diameter; ya 10th dark ring se lens ki radius of curvature.

4. □ (a) Michelson interferometer ki construction & working; mirror displacement se fringe shift ka expression.

□ (b) Rayleigh criterion of resolution — resolve karne ki condition samjhao.

5. □ (a) Plane transmission grating ki theory; resolving power ka expression aur dispersive power se relation.

□ (b) Numerical: 4250 lines/cm grating, 2nd order line 30° par → sodium light ka wavelength.

6. □ (a) Fraunhofer diffraction at a single slit — intensity distribution; first subsidiary maximum ≈4.5% show karo.

□ (b) Write short note: physical significance of wave function (Born interpretation).

7. □ (a) Optical fibre ke liye acceptance angle aur numerical aperture (NA) derive karo.

□ (b) Numerical: core 1.55, cladding 1.50 → NA, acceptance angle, critical angle, acceptance cone.

8. □ Short notes (any two): (a) de Broglie matter waves (b) Compton shift (c) Fresnel's biprism interference (d) Mach–Zehnder interferometer.

Predicted Paper — BT-202 Mathematics-II (Mid-Sem / MSE-1)

Jun2025→Jun2026 ke 3 papers + earlier sessions · attempt any five · 14 marks each

1. □ (a) Linear differential equation solve karo (integrating factor method). Type: $dy/dx + P(x)y = Q(x)$.

□ (b) Bernoulli's equation solve karo (y^2 wale term ke saath, $v = 1/y$ substitution).

2. □ (a) Homogeneous / variable-separable first-order equation solve karo.

□ (b) Solve $(D^2 - 5D + 6)y = 4e^x + 5$ type — CF + PI (operator method).

3. □ (a) Method of variation of parameters se solve karo ($y'' + y = \tan x$ / $\operatorname{cosec} x$ ya $(D^2 + a^2)y = \tan ax$).

□ (b) Cauchy–Euler / reducible-to-homogeneous equation: $(1+x)^2y'' + (1+x)y' + y = f(\log(1+x))$.

4. □ (a) Frobenius method se series solution nikaalo ($xy'' + y' + xy = 0$ about $x = 0$ type).

□ (b) Legendre's differential equation ko series me solve karo.

5. □ (a) Simultaneous differential equations (dx/dt, dy/dt wale) solve karo.

□ (b) Bessel function identity prove karo: $J_{1/2}(x) = \sqrt{(2/\pi x)}\cdot\sin x$.

6. □ (a) Exact differential equation / integrating factor se solve karo.

□ (b) First order but not first degree — solvable for p/x/y ya Clairaut's form.

7. □ (a) Orthogonal trajectories / application based first-order ODE.

□ (b) Total differential equation solve karo.

8. □ Short notes: (a) Wronskian & linear independence (b) Complementary function vs particular integral (c) Laplace transform se ODE solve karna (agar syllabus me ho) (d) Legendre polynomials ke basic properties.

Predicted Paper — BT-203 Basic Mechanical Engineering (Mid-Sem / MSE-1)

Nov2022→Jun2026 ke 4 papers · attempt any five · 14 marks each

1. □ (a) Iron–Carbon ($Fe–Fe_3C$) equilibrium diagram draw karo — saare phases, invariant reactions aur critical temperatures explain karo.

□ (b) Cast iron ki composition, classification aur properties describe karo.

2. □ (a) Stress–strain curve of ductile material — yield point aur ultimate tensile strength ka significance. Numerical bhi aa sakta hai (rod, axial pull, E).

□ (b) Mechanical properties define karo: strength, hardness, toughness, ductility, brittleness, malleability.

3. □ (a) Lathe machine ke main parts neat diagram ke saath; koi do operations detail me explain karo.

□ (b) Lathe par ki jaane wali operations ki list + vernier calliper se measurement (least count).

4. □ (a) Temperature measuring instruments ki list + optical pyrometer ka working principle (diagram).

□ (b) Measurement me errors ke types; ya slip gauge / sine-bar / dial gauge ke short notes.

5. □ (a) Hooke's law derive karo aur modulus of elasticity ka concept.

□ (b) Classification of engineering materials; alloys kya hain + applications.

6. □ (a) Viscosity define karo + numerical (flat plate, gap, speed, poise → force & power).

□ (b) Newton's law of viscosity / homogeneous vs heterogeneous fluids.

7. □ (a) Hardness measure karne ke methods.

□ (b) Alloy steel aur alloying elements ka steel properties par effect.

8. □ Short notes (any two): (a) Fe–Fe_3C me critical temperatures (b) Welding defects (c) Dial gauge (d) Manometer & pressure types.

Predicted Paper — BT-204 Basic Civil Engg. & Mechanics (Mid-Sem / MSE-1)

Nov2022→Jun2026 ke 3 papers · attempt any five · 14 marks each

1. □ (a) Cement ke properties, laboratory tests aur engineering applications; concrete ki quality par effect.

□ (b) Bricks aur stones ke types, properties aur uses describe karo.

2. □ (a) Stones ki characteristics + testing methods; building work me use hone wale stones ke naam.

□ (b) Stone masonry ke types typical sketches ke saath.

3. □ (a) Levelling numerical — Height of Instrument ya Rise & Fall method se RL nikaalo + arithmetic check.

□ (b) Height of Collimation vs Rise and Fall method ka difference.

4. □ (a) Theodolite ke individual components sketch ke saath; principle & applications.

□ (b) Remote sensing ke applications (terrain analysis, site investigation, material inventories).

5. □ (a) Contouring ka principle; contour line, contour interval, horizontal equivalent, contour map ke uses.

□ (b) Spread footing aur RCC footing — construction & suitability neat sketch ke saath.

6. □ (a) GPS ke saath positioning ka principle (neat sketch).

□ (b) Distances & elevations measurement ke methods; conventional vs EDM comparison.

7. □ (a) Mortar kya hai + requirement estimation; brick masonry ke general principles.

□ (b) Staircase / windows / doors ke types (sketch).

8. □ Short notes (any two): (a) Plastering & pointing (b) Plane table & instruments (c) Survey stations classification (d) Area & volume computation.

Predicted Paper — BT-205 Basic Computer Engineering (Mid-Sem / MSE-1)

Jun2020→Jun2026 ke 6 papers · attempt any five · 14 marks each

1. □ (a) Operating system — definition, major functions; file, process aur memory management detail me explain karo.

□ (b) System software vs application software (examples ke saath).

2. □ (a) Computer ka block diagram / CPU organization — registers, bus architecture, instruction set.

□ (b) Computer ki classification (analog/digital/hybrid) aur generations.

3. □ (a) Memory & storage systems + I/O devices / memory hierarchy.

□ (b) MS-Word, MS-Excel, MS-PowerPoint ke features + office automation me importance.

4. □ (a) Algorithm define karo + characteristics; three numbers me se largest nikaalne ka flowchart.

□ (b) Flowchart ke advantages / program design.

5. □ (a) OOP ke fundamental characteristics aur POP ki limitations kaise overcome hoti hain.

□ (b) POP vs OOP comparison table.

6. □ (a) C++ program ki basic structure aur tokens (diagram/program ke saath).

□ (b) Arrays (1-D vs multi-dimensional) aur different types of functions.

7. □ (a) Inheritance in C++ — types with diagrams; single vs multiple inheritance.

□ (b) Operator overloading / function overloading with example; virtual functions.

8. □ Short notes (any two): (a) Scope resolution operator (b) Constructors & destructors (c) Friend function (d) Looping statements with example program.

End-Sem (Unit 3-5) — Most-probable topics

BT-201 — Engineering Physics

□ **U3** Hall effect + Hall voltage derivation & applications
□ **U3** Fermi energy & Fermi level (intrinsic/extrinsic); density of states; Bloch's theorem
□ **U3** P-N junction diode I–V characteristics; Zener diode; solar cell
□ **U4** Einstein coefficients (A & B) relation + population inversion
□ **U4** He–Ne laser / CO_2 laser — construction & working
□ **U4** Optical fibre — NA, acceptance angle, losses (V-number)
□ **U5** Maxwell's equations in vacuum/free space
□ **U5** Poynting theorem + physical significance
□ **U5** Gauss divergence theorem / continuity equation
□ **U5** Electric field & potential due to continuous charge distribution

BT-202 — Mathematics-II

□ **U3** PDE formation, Lagrange's method $(y+z)p+(x+z)q=x+y$ type)
□ **U3** Charpit's method / complete integral
□ **U3** Second order PDE $(D^2-2DD'+D'^2)Z = f(x,y)$ solve karna

 U4 Harmonic function + harmonic conjugate nikalna
 U4 Analytic function ki condition (find K value)
 U4 Cauchy's integral theorem & integral formula
 U4 Residues & poles; residue theorem se integral
 U5 Directional derivative numerical
 U5 Gradient, divergence, curl (definitions + numerical)
 U5 Stokes'/Green's theorem verify karna
 U5 Line integral / work done by force

BT-203 — Basic Mechanical Engineering

 U3 Bernoulli's equation derivation + assumptions + physical significance
 U3 Viscosity / Newton's law; manometer & pressure types
 U3 Pumps — classification, working principle (reciprocating/centrifugal)
 U3 Hydraulic turbines — classification, impulse vs reaction, Pelton wheel
 U4 Zeroth / second law of thermodynamics (Clausius & Kelvin–Planck)
 U4 Carnot cycle + efficiency derivation
 U4 Otto & Diesel cycle + efficiency
 U4 Four-stroke / two-stroke petrol & diesel engine
 U4 Boiler — classification, draught, boiler trial numerical
 U5 Compressor — definition, working principle (reciprocating)
 U5 Thermodynamic system types & processes (constant P/V)

BT-204 — Basic Civil Engg. & Mechanics

 U3 Contouring, profile cross-sectioning, computation of areas & volumes
 U3 Remote sensing & GPS applications
 U4 Free body diagram, equilibrium conditions of coplanar forces
 U4 Lami's theorem + numerical
 U4 Analysis of plane trusses — Method of joints & Method of sections (+ difference)
 U4 Friction ke laws + inclined plane numerical (block on incline)
 U5 Centroid & centre of gravity (difference + numerical)
 U5 Moment of inertia vs product of inertia; radius of gyration
 U5 SFD & BMD — simply supported beam (point load, UDL, couple)
 U5 Support reactions / work-energy principle / D'Alembert's principle

BT-205 — Basic Computer Engineering

 U3 Object & class, structure of class, class vs structure; inheritance programs
 U3 Constructor, destructor, function & operator overloading programs
 U3 C++ programs: stack insert/delete, binary→decimal, loops
 U3 Data structures intro — linear vs nonlinear, arrays/stack/queue
 U4 ISO-OSI reference model — layers & functions
 U4 TCP/IP model + internetworking devices
 U4 Computer security — viruses, worms, Trojans, malware, phishing, DoS
 U4 Firewall, computer ethics & cyber laws, E-commerce
 U5 DBMS — advantages over file system, DDL/DML, primary key, DBA
 U5 Database transaction & data integrity; data dictionary; data models
 U5 Cloud computing — models, IaaS/PaaS/SaaS, types, advantages

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Mid-sem papers college-level internal hote hain (public nahi) — ye prediction RGPV ke end-sem papers ke repeat-pattern par based hai, isliye probability 'chances' batati hai, guarantee nahi. Exam se pehle apne college ka syllabus/circular verify karein. Papers RGPV ki property hain — free study use only.